

INTRODUCTION TO COMPUTER SYSTEM, NETWORKS AND SECURITY

Assignment 4 – Cybersecurity Part 1 – Reflection

Submitted by;
Rahul Vedant Pemsing
36305529
rpem0003@student.monash.edu

MONASH University
School of Information Technology
FIT1047

Purpose for the whole assignment 4:

- ❖ Students will analyse and discuss a recent vulnerability or cybersecurity attack. This demonstrates an understanding of related cybersecurity topics and the ability to research information on cybersecurity incidents.
- ❖ Students will write a report on how a specified set of security controls is implemented in a medium-sized enterprise scenario. This report shows the understanding of various security controls in the departments and the ability to evaluate and explain their utilisation.
- ❖ Students will examine the security of two or more apps in the same category on their mobile device, such as Entertainment, Shopping, Education, Messaging, Banking, Games, or Social Networks, showing an understanding of mobile security, privacy and ethical concerns.

Purpose for the assignment 4 part 1:

TASK	BRIEF DESCRIPTION
Reflection	Reflect on what was done during week 10 to week 12 of classes; what you learned, what was difficult...

FIT1047 Weekly Reflection

Week 10:- Cybersecurity Basics

Key Learnings:

- Describe basic security mechanisms (encryption, digital signatures, authentication)
- Discuss security properties achieved by applying different types of cryptography
- Explore application scenarios of authentication mechanisms
- Describe access controls in the context of computer architecture

Personal Insights:

- I learned how essential cryptography is in ensuring data confidentiality, integrity, and authenticity. Understanding the role of encryption and digital signatures gave me a clearer picture of how secure communications are established in real-world systems.

Application of Knowledge:

- I applied these concepts by analysing how public and private keys are used in SSL/TLS connections and how authentication mechanisms like passwords and biometrics protect system access.

—Assignment 2— MARIE Programming

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- I also applied my knowledge in the applied classes and in the test in week 12.

The screenshot shows a presentation slide from Monash University. The top section is titled 'Activity 2: RSA Algorithm' and contains four numbered steps: 1. Server generated public key (7, 33) and private key (3, 33). 2. Browser requests the Server's public key (7, 33). 3. Browser encrypts plain message, M=14 into encrypted message, C=20 and sends it to the Server. 4. Server decrypts the encrypted message into its original plain message. Below these steps are three tasks: Task 1: Generating public and private keys (with handwritten calculations for p=3, q=11, phi=20, e=3, d=3), Task 2: Encrypting the message (C = M^e mod n = 14^3 mod 33 = 20), and Task 3: Decrypting the message (M = C^d mod n = 20^3 mod 33 = 14). The middle section is titled 'Recap: Diffie-Hellman Key Exchange' and includes the DHKE steps: 1. Alice and Bob agree on a generator, g and a large prime integer, p. 2. Choose a random private key, X_A and X_B, that less than p. 3. Compute public key, Y_A and Y_B. 4. Exchange public key. 5. Compute common secret key. Handwritten calculations show Alice's X_A=6, Y_A=7^6 mod 11=4, and Bob's X_B=9, Y_B=7^9 mod 11=8. The common secret key K_{AB}=4^9 mod 11=3. The bottom section is titled 'Activity 3: DHKE' and describes a scenario where Alice (A) and Bob (B) wish to do DHKE with p=47 and g=5. Alice's private exponent is a=3, and Bob's is b=7. Handwritten calculations show Y_A=5^3 mod 47=31, Y_B=5^7 mod 47=11, and the common secret key K_{AB}=31^7 mod 47=15. A diagram illustrates the process using colored circles representing public and private keys and the resulting common secret.

Challenges Faced:

- The most challenging part was distinguishing between symmetric and asymmetric encryption in terms of use cases and key management.

Future Goals:

- I aim to strengthen my understanding of encryption algorithms and practice using tools such as Wireshark or OpenSSL to visualize how secure data transmission occurs.

Week 11:- Network Security

Key Learnings:

- Describe key establishment mechanisms and transport layer security protocols
- Discuss the use of firewalls and VPNs in different network scenarios
- Understand how large and small organisations can use security controls to provide perimeter protection for their networks
- Identify the use of security protocols in network traffic

Personal Insights:

- This week enhanced my understanding of how data travels securely over networks. Learning about VPNs and firewalls helped me appreciate how organisations control and protect network traffic.

Application of Knowledge:

- I practiced identifying secure communication protocols and simulated basic firewall rules in applied exercises to understand packet filtering and network segmentation.
- I also applied my knowledge in the assignment 4

Challenges Faced:

- Configuring and understanding VPN tunnelling concepts was initially confusing, especially in differentiating between site-to-site and remote-access VPNs.

Future Goals:

- I plan to gain more hands-on experience by setting up a small virtual network with firewalls and VPNs to deepen my practical skills.

Week 12:- Vulnerabilities and Attacks

Key Learnings:

- Explain cybersecurity properties
- Describe different types of malware and cybersecurity attacks
- Understand sources of vulnerabilities
- Understand the risk of cyberattacks to connected systems
- Explore mechanisms to find devices connected to the Internet
- Discuss privacy-preserving mechanisms and advantages/disadvantages of these mechanisms

Personal Insights:

- This week was eye-opening as I learned how attackers exploit system vulnerabilities and the real-world impact of malware and phishing. It made me more aware of the importance of constant patching and user awareness in cybersecurity.

Application of Knowledge:

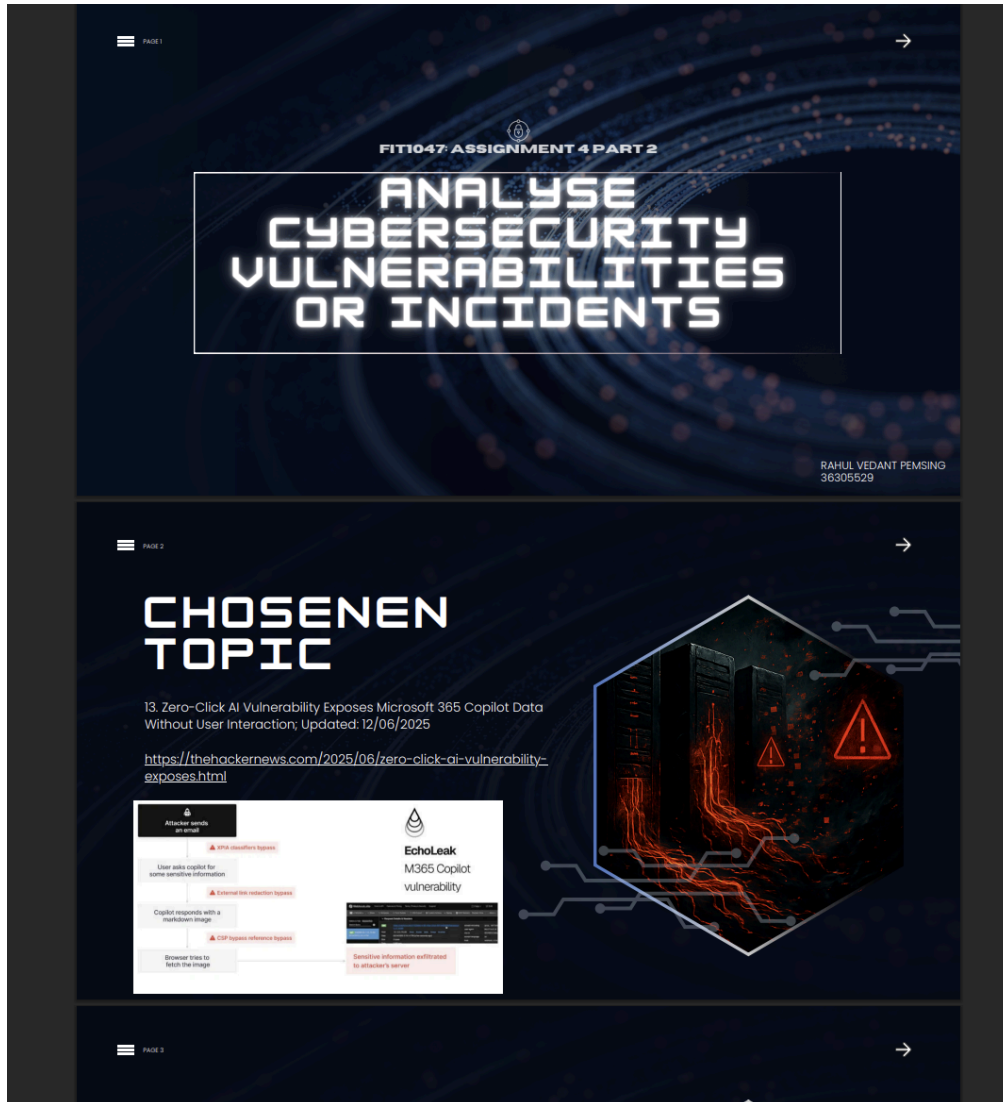
- I researched recent cyberattacks and analysed how attackers exploited software flaws, which helped me connect classroom theories to real-world incidents.
- I applied my knowledge in the applied class.

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- I applied that knowledge in assignment 4.



Challenges Faced:

- It was challenging to fully understand how zero-day vulnerabilities are detected and mitigated due to the complexity of modern systems.

Future Goals:

- I intend to study penetration testing techniques and vulnerability scanning tools to better identify and mitigate system weaknesses.